

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 422 — Past Questions & Materials
by Renance

Course: MTH 422
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

This study pack was compiled and restructured by Renance for MTH 422 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — MTH 422 Lebesgue Measure and Integration

Second Semester 2020 2021

Level 400 • Second Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
SECOND SEMESTER EXAMINATION 2020/2021 SESSION
COURSE TITLE & CODE: LEBESQUE MEASURE AND INTEGRATION (MTH 422)

Instruction: Attempt any four questions Time: 2.5 hours

Question 1 (17.5 marks)

- Define the following terms (i) A δ field (ii) A Borel Algebra (iii) A measurable space (12.5 marks)
- Let $A_1, A_2, \dots, A_n$ be pairwise disjoint countable measurable set functions. Then prove that $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$, $\mu(A_n) < \infty, \forall n$. (5 marks)

Question 2 (17.5 marks)

- Let (X, Ω) be a measurable space and let X, Y be topological spaces such that $f: X \rightarrow Y$ is measurable and $g: Y \rightarrow Z$ is continuous then prove $g \circ f: X \rightarrow Z$ is also a measurable function. (5 marks)
- Let (X, Ω) be a measurable space and let $U: X \rightarrow \mathbb{R}, V: X \rightarrow \mathbb{R}$ be measurable functions on X . Let $\varphi: \mathbb{R}^2 \rightarrow Y$ be a continuous then prove that $h: X \rightarrow Y$ defined by $h(x) = \varphi[U(x), V(x)]$ is also a measurable space. (6 marks)
- If g and f are Lebesgue integrable functions then prove that $\int_x (\alpha f + \beta g) d\mu = \alpha \int_x f d\mu + \beta \int_x g d\mu$ where α and β are elements in the complex field. (6.5 marks)

Question 3 (17.5 marks)

- Let f be a measurable function on X , then prove that there exists a complex valued function on $X \ni |\alpha| = 1$ and $f = \alpha \times |f|$. (7.5 marks)
- Let X be a non-empty set, a subset of X is called a σ -algebra i.e. $M \subset P(X)$. A measure function is a map $\mu: P(X) \rightarrow [0, \infty]$ that satisfies the following axioms: (i) $X \in M$ (ii) if $A_i \in M$, then $A^c \equiv X: A \in M$ (iii) if $A_i \in M$, then $\bigcup_{i=1}^{\infty} A_i \in M$ (10 marks)

Question 4 (17.5 marks)

Let (X, M_0) be a measurable space and let (Y, τ) be a topological space. Let $f: X \rightarrow Y$ be a function then prove that

- $\Omega = \{E \subset Y: f^{-1}(E) \in M_0\}$ is a σ -(sigma) algebra in Y . (5 marks)
- If f is measurable and E is a Borel set in Y , then prove that $f^{-1}(E) \in M_0$. (5 marks)
- If f is measurable and $g: Y \rightarrow Z$ is a Borel measurable function, then prove that $g \circ f$ is measurable. (5½ marks)

Question 5 (17.5 marks)

- Let X be any non-empty set and $M_0 = \mathbb{R}(X)$. Let μ be defined by $\mu(E) = \begin{cases} n(E), & \text{if } E \text{ is finite} \\ \infty, & \text{if } E \text{ is infinite} \end{cases}$ show that μ is a measure on X or $(X, \mathbb{R}(X), \mu)$ is a measure space. (5.5 marks)
- Let $\{A_n\}$ be a countable collection of sets of real numbers, then $m^*(\bigcup_{i=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} m^*(A_n)$. Prove. (6 marks)
- Let A be any sets and $E_1, E_2, \dots, E_n$ be a finite sequence of pairwise disjoint $m^*[A \cup (\bigcup_{i=1}^{\infty} E_i)] = \sum_{i=1}^{\infty} m^*(A \cap E_i)$. Prove. (6 marks)

Question 6 (17.5 marks)

- Let A be any sets and $E_1, E_2, \dots, E_n$ be a finite sequence of pairwise disjoint $m^*[A \cup (\bigcup_{i=1}^{\infty} E_i)] = \sum_{i=1}^{\infty} m^*(A \cap E_i)$. Prove. (7.5 marks)
- Prove the Levi's Theorem as stated. Suppose $f_1(x) \leq f_2(x) \leq \dots \leq f_n \leq M$ are monotone functions on a set A where the functions are all integrable and $\int_A f_n d\mu \leq M$; $(1, 2, \dots)$, for some constants M . Then the $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ exists almost everywhere on A and moreover f is integrable and $\lim_{n \rightarrow \infty} \int_A f_n d\mu = \int_A f d\mu$. (10 marks)